

Statistical and Machine Learning

Tree based methods and boosting

Instructions

All questions should first be discussed in pairs or small groups. Afterwards, we shall discuss them together. Solutions will be distributed after each session.

Exercise 1 — Decision Tree Terminology

Consider a decision tree fitted to predict baseball players' salaries using two predictors: **Years** and **Hits**. The tree produces three terminal regions:

$$R_1 = \{X \mid \text{Years} < 4.5\}, \quad R_2 = \{X \mid \text{Years} \geq 4.5, \text{ Hits} < 117.5\}, \quad R_3 = \{X \mid \text{Years} \geq 4.5, \text{ Hits} \geq 117.5\}.$$

- (a) Identify the **root node**, one **internal node**, and the **terminal nodes** (leaves).
 - (b) A new player has 6 years of experience and 130 hits. Which region does this player fall into? If the average log-salary in that region is $\hat{c}_m = 6.74$, what is the predicted salary?
 - (c) Explain the difference between how a **regression tree** and a **classification tree** make predictions in a terminal node.
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Exercise 2 — CART Split Selection

Fit a regression tree to a node with 6 observations: $y = (1, 3, 3, 5, 8, 10)$, $X = (2, 4, 5, 7, 9, 11)$.

Consider two candidate splits: $X < 5.5$ and $X < 8$.

- (a) For each split, identify the left and right regions and compute region means $\hat{c}_1$, $\hat{c}_2$.
 - (b) Compute $\text{RSS}(j, s) = \sum_{i: x_i < s} (y_i - \hat{c}_1)^2 + \sum_{i: x_i \geq s} (y_i - \hat{c}_2)^2$ for each split.
 - (c) Which split does CART select?
 - (d) Does CART evaluate all real-valued split points? How is the search made computationally feasible?
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Exercise 3 — Impurity Measures for Classification Trees

A binary split produces:

- **Left** R_1 : 40 observations — 30 class 1, 10 class 0.
- **Right** R_2 : 60 observations — 10 class 1, 50 class 0.

- (a) Compute the proportion of class 1, $\hat{\pi}_{m1}$, in each node.
- (b) Compute for each node: misclassification error $i_1 = 1 - \max_k \hat{\pi}_{mk}$, Gini index $i_2 = 1 - \sum_k \hat{\pi}_{mk}^2$, and entropy $i_3 = -\sum_k \hat{\pi}_{mk} \log(\hat{\pi}_{mk})$.

- (c) Compute the weighted Gini impurity $i_w = \frac{40}{100}i_2(R_1) + \frac{60}{100}i_2(R_2)$.
- (d) Why is misclassification error often a less useful criterion for guiding splits than Gini or entropy?
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Exercise 4 — Bagging and Out-of-Bag Error

- (a) Describe the two main steps of bagging: bootstrap and aggregating.
- (b) A training set has $m = 200$ observations. A bootstrap sample is drawn with replacement.
- Probability of a given observation **not** being selected in one draw?
 - Approximate probability of being **absent** from the entire bootstrap sample?
 - Expected number of **OOB** observations?
- (c) How can OOB observations estimate the test error without a separate validation set?
- (d) Bagging assigns equal weights $\beta_j = 1/M$ to all trees. How does this differ from boosting?
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Exercise 5 — Random Forests vs. Bagging

- (a) What is the key algorithmic difference between random forests and standard bagging? Why is this modification introduced?
- (b) For a classification problem with $p = 100$ predictors, what is the typical recommended `mtry`?
- (c) Using $\text{Var}(\frac{1}{M} \sum X_m) = \rho\sigma^2 + \frac{1-\rho}{M}\sigma^2$, explain why decorrelating trees reduces ensemble variance.
- (d) In an ensemble of $M = 3$ independent classifiers each correct with probability 0.6, compute the probability the majority vote is correct. What if classifiers are perfectly correlated?
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Exercise 6 — Boosting: Residuals, AdaBoost, and Gradient Boosting

- (a) In boosting for regression trees, successive trees are fitted to **residuals**. Explain the intuition.
- (b) Boosting has three tuning parameters: number of trees B , shrinkage λ , and splits per tree d . Describe the role and trade-offs of each.
- (c) In AdaBoost, $\alpha_m = \log\left(\frac{1-\text{err}_m}{\text{err}_m}\right)$. Compute $\alpha_1, \alpha_2, \alpha_3$ for $\text{err} = 0.1, 0.3, 0.45$ and interpret.
- (d) Explain the analogy between **gradient boosting** and gradient descent. What are the pseudo-residuals?
- (e) XGBoost uses $\Omega(f) = \gamma J + \frac{1}{2}\lambda \sum_{j=1}^J w_j^2$. Explain the role of γ , λ , J , and w_j .